

Name _____ Date _____

Unit 3B: Rational Number Operations, Day 7: Multiplying by Proper and Improper Fractions

	Second Factor Greater Than or Less Than One?	Product	Was the First Factor Increased or Decreased?
$\frac{1}{2} \times \frac{3}{4}$	less	$\frac{3}{8}$	decreased
$\frac{3}{5} \times \frac{3}{5}$	less	$\frac{9}{5} = 1\frac{4}{5}$	decreased
$\frac{1}{5} \times \frac{3}{2}$	More	$\frac{3}{10}$	increased
$\frac{2}{1} \times \frac{7}{4}$	More	$\frac{14}{4} = 3\frac{1}{2}$	increased
$\frac{2}{5} \times \frac{1}{2}$	less	$\frac{2}{10} = \frac{1}{5}$	decreased
$\frac{3}{8} \times \frac{4}{3}$	More	$\frac{12}{24} = \frac{1}{2}$	increased

What happens when you multiply a quantity by a **proper** fraction

(Less than one)? The product decreases (Answer gets smaller)

What happens when you multiply a quantity by an **improper** fraction?

(Greater than one)? The product increases (Answer gets bigger)

1. Mr. Tran used $\frac{7}{8}$ of a box of nails to build a dog house. The box of nails weighs 20 ounces. Which statement is true about the weight of the nails Mr. Tran used?

A. He used less than 20 ounces of nails.

$$20 \times \frac{7}{8}$$

B. He used exactly 20 ounces of nails.

C. He used more than 20 ounces of nails.

D. Not Here

Justify your answer in the space below.

He used part of the box.
 $\frac{7}{8}$ is a proper fraction less than 1
 $20 \cdot \frac{7}{8}$ would be less than 20.

2. On Monday, Sharonda spent 36 minutes practicing her flute solo. On Tuesday, she practiced for $\frac{11}{12}$ of Monday's time. On Wednesday, she practiced for $\frac{14}{12}$ of Monday's time. Which of the following statements most accurately compares the time Sharonda spent practicing?

Tues 36 $\cdot \frac{11}{12}$

Tuesday:

Wednesday: Wed 36 $\cdot \frac{14}{12}$

- ✓ I. Sharonda spent less than 36 minutes practicing on Tuesday.
- II. Sharonda spent ~~more~~ than 36 minutes practicing on Tuesday.
- III. Sharonda spent ~~less~~ than 36 minutes practicing on Wednesday.
- ✓ IV. Sharonda spent more than 36 minutes practicing on Wednesday.

A. I and II only

B. I and IV only

C. II and III only

D. II and IV only

Justify your answer in the space below.

Tuesday is $\frac{11}{12}$ times 36. This proper fraction will cause the answer to get smaller.
 $\frac{14}{12}$ (improper) will cause the answer to get larger.

3. Mr. Thorn gave his son a math riddle to solve. He said, "When you multiply a whole number by this number, it will increase the value." Which values could represent Mr. Thorn's number?

improper

I. $\frac{5}{8}$

II. $\frac{3}{2}$

III. $\frac{4}{5}$

IV. $\frac{7}{3}$

A. I and III only

B. I and IV only

C. II and III only

D. II and IV only

Justify your answer in the space below.

When multiplying by improper fractions, the "value" will increase.